



5 CONSERVATION

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Kissimmee

2040 Comprehensive Plan

Conservation

prepared for

City of Kissimmee

prepared by

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date

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Cover photo source: City of Kissimmee Public Information Office

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GOALS, OBJECTIVES, AND POLICIES

This section stipulates goals, objectives, and implementing policies for the Conservation Element pursuant to §163.3177(6)(d), Florida Statutes.

GOAL 5.1: CONSERVATION.

Conserve, protect, and appropriately manage the natural resources within the corporate limits of the City in order to enhance the quality of natural systems, including Lake Tohopekaliga, Shingle Creek, and Mill Slough, as well as associated floodplains, wetlands, flora, and fauna, the surficial and Floridian aquifers, and other natural resources, as below directed.

OBJECTIVE 5.1.1: PROTECT AIR QUALITY.

The City shall meet or exceed the minimum air quality standards established by the Florida Department of Environmental Protection (FDEP).

POLICY 5.1.1.1: AIR POLLUTION AND LAND USE REGULATIONS.

The City shall continue to protect against loss of air quality by maintaining land use controls that promote only activities compatible with existing land uses and natural systems and prohibit activities that generate air pollutants and other adverse impacts. The City's Land Development Code shall continue to require that air pollutants (smoke, certain greenhouse gases, particulate matter, odor, and toxic matter) comply with FDEP standards.

POLICY 5.1.1.2: AIR POLLUTION MONITORING PROCEDURES.

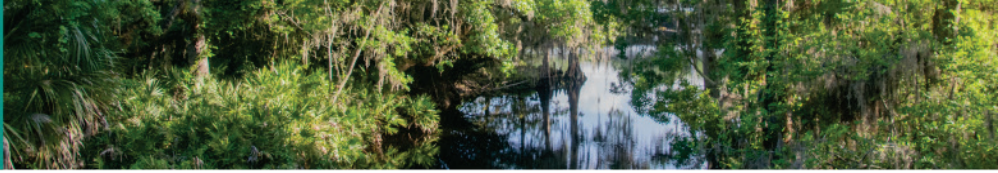
The City shall continue to work with FDEP, MetroPlan Orlando, and other organizations to improve air quality in the region.

POLICY 5.1.1.3: COMBAT EROSION AND GENERATION OF DUST PARTICLES.

Land Development regulations shall maintain performance standards that combat erosion and generation of fugitive dust particles. The regulations shall require that measures be taken on building sites or cleared areas which assure that exposed, destabilized, or other altered soil is expeditiously covered with an acceptable erosion control material, per construction and development best management practices.

OBJECTIVE 5.1.2: WATER QUALITY.

The water quality regulations established by the South Florida Water Management District (SFWMD) shall be enforced by the City, in order to protect the quality of current and projected



future water sources and surface water run-off. The City shall maintain water quality regulations in accordance with Chapter 4A: Potable Water Sub-Element of the Public Facilities Element.

POLICY 5.1.2.1: PROTECTION OF CLASS III WATER BODIES.

Class III Waters are waterbodies that currently support recreation and foster the maintenance of fish and aquatic wildlife. All lakes and major tributaries within the City of Kissimmee, including Lake Tohopekaliga, Shingle Creek, and Mill Slough, are Class III waters. These waters shall be protected by maintaining the following provisions in the City's land development regulations:

1. Dredging activities shall be limited to Florida Department of Environmental Protection (FDEP) approved dredging.
2. Ensure water quality by coordinating with the FDEP and the SFWMD in maintaining records of stormwater management practices, usage, construction and maintenance costs, and any required monitoring of selected stormwater management facilities. The City shall notify the appropriate agency with jurisdiction as potential issues or problems are identified by the City.
3. Limit the use of Class III waters to water dependent activities that are not contrary to the public interest and satisfy a community need.
4. Where modification of grassbeds is permitted the City shall assure that:
 - a. A determination of overriding public interest has been demonstrated by the applicant prior to modification of grassbeds.
 - b. Project run-off, including nitrogen, phosphorous, and other nutrients, shall be controlled to prevent an increase in water turbidity, decrease in fish or aquatic wildlife propagation, or other degradation of water quality.
 - c. Projects damaging grassbeds during construction shall incorporate mitigative techniques that re-establish conditions favorable to natural regeneration of the grassbeds.
 - d. Special attention shall be given to stipulations cited in (1) through (3) above during the development review process. The City shall coordinate closely with State and federal agencies during the permitting processes to insure that the intent of these policies is carried out.

POLICY 5.1.2.2: COORDINATION WITH WATER RESOURCE AGENCIES.

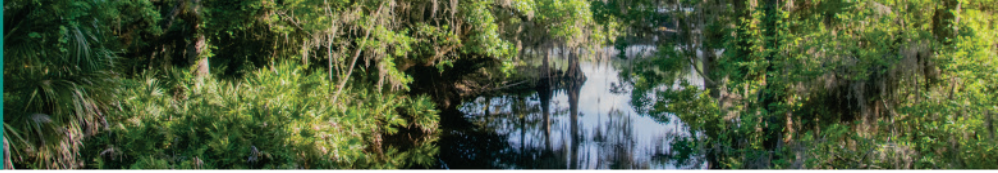
The City of Kissimmee shall continue to coordinate with appropriate agencies to protect the quality of the City's water resources, including:

1. The City of Kissimmee shall continue to cooperate with the Florida Fish and Wildlife Conservation Commission's (FWWCC) Lake Tohopekaliga fisheries management efforts, which includes a lake restoration program. The restoration program involves periodic drawdowns of lake waters for removal of organic sediment.
2. The City of Kissimmee shall continue to cooperate with the U.S. Army Corps of Engineers which serves as the lead agency in an ongoing water quality enhancement program impacting Shingle Creek.
3. The City of Kissimmee shall continue to promote land acquisition programs to purchase property rights in the vicinity of portions of Shingle Creek, and Reedy Creek within and nearby Kissimmee for the purpose of conserving and protecting environmentally sensitive lands. Purchase of this property is intended to enhance floodplain protection, water management, water supply, and water resource protection efforts, while allowing for appropriate recreation and open space uses.
4. The City shall also continue to coordinate with the East Central Florida Regional Planning Council (ECFRPC) and the FDEP in matters regarding documentation of:
 - a. Stormwater best management practices;
 - b. Stormwater facility construction and maintenance costs; and
 - c. Monitoring of selected stormwater management facilities.
5. The City shall continue to coordinate with the ECFRPC, FDEP, the SFWMD and other entities to educate the general public on the nonpoint source pollution problems and available stormwater management practices to abate nonpoint source impacts on water quality.

POLICY 5.1.2.3: REGULATE AGRICULTURAL ACTIVITIES TO PRESERVE WATER QUALITY.

The City shall continue to ensure that any agricultural activities shall:

1. Not be conducted adjacent to existing waterways and surface water management activities and shall comply with all applicable policies of the FDEP, SFWMD, and other agencies having appropriate jurisdiction as well as State laws;
2. Maintain natural drainage patterns;



- a. Promote the use of alternative surface water supplies for irrigation purposes;
- b. Prohibit the expansion of agricultural activities into wetland areas; and
- c. Use best management principles and practices in order to reduce pesticide and fertilizer run-off, prevent soil erosion, and preserve water quality.

POLICY 5.1.2.4: REGULATE WASTEWATER TREATMENT DISCHARGE TO PRESERVE WATER QUALITY.

The City shall continue to implement the following performance standards in order to protect water quality:

1. All new residential subdivisions as well as multiple family and non-residential development within the City of Kissimmee shall be required to connect to the public wastewater system pursuant to Chapter 10-5 of the Code of Kissimmee.
2. In areas where developments cannot be connected to the public wastewater system, private wastewater disposal systems are acceptable as interim measures providing such facilities are approved by the City Water Resources Director's office. Notwithstanding, all private wastewater disposal systems shall be designed to facilitate mandatory hookups to the public wastewater system when the public system becomes available.
3. The City of Kissimmee shall promote application of innovative concepts in wastewater collection and disposal including wastewater reuse through such programs as use of "graywater" for spray irrigation and other non-potable uses.

POLICY 5.1.2.5: SHORELINE PRESERVATION.

The City shall continue to require that all new development preserve native vegetation and re-vegetate areas of the shorelines of Lake Tohopekaliga, Shingle Creek, Mill Slough, and the City Ditch, where erosive tendencies are present, with native species to reduce erosion.

POLICY 5.1.2.6: SHORELINE SETBACKS.

In addition, the City of Kissimmee shall continue to protect the floodplain of Shingle Creek, the East and West City Ditch, and Mill Slough by requiring that any development maintain a minimum 150 foot setback from the edge of these riparian and drainage waterbodies. The City Engineer shall maintain authority to require greater setback for purposes of protecting floodplains and other environmentally sensitive areas which require a greater setback for purposes of floodplain protection and preservation.

POLICY 5.1.2.7: PROTECT SURFICIAL AQUIFER RECHARGE AREAS.

The City of Kissimmee shall continue to require the incorporation of open space and/or pervious surfaces in order to preserve the quality of water resources within the surficial aquifer.

POLICY 5.1.2.8: WELLFIELD PROTECTION.

The City shall continue to enforce the city's wellfield protection ordinance, as amended from time to time, to prevent future intrusion of groundwater contaminants into the Floridian Aquifer wellfield areas, create buffer zones around wellfields, and controlling land uses around wells.

POLICY 5.1.2.9: LOW-IMPACT DEVELOPMENT TECHNIQUES.

The City shall continue to encourage the use of low-impact development techniques for on-site stormwater quality improvements for all land development and redevelopment projects.

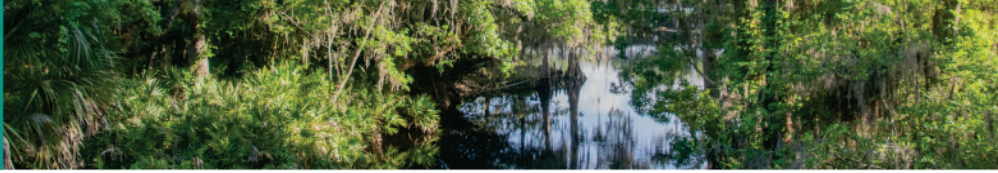
OBJECTIVE 5.1.3: WATER QUANTITY.

The City shall continue to maintain its adopted level of service standard in order to conserve the availability of current and future water sources.

POLICY 5.1.3.1: CONSERVATION OF POTABLE WATER SUPPLY.

In order to conserve the potable water supply and to achieve a reduction in the current rates of water consumption, the City shall continue to enforce the following performance standards:

1. Where non-potable alternative sources of irrigation water are available, potable water supplies may not be used to meet irrigation needs.
2. Encourage adherence to water saving plumbing fixtures in all new construction or renovations. Also, the City of Kissimmee shall conserve water supply through use of a leak detection program designed to curtail wasteful loss of potable water resources.
3. In order to reduce demand for irrigation water, new development shall employ and/or preserve native vegetation, or use other drought resistant plants for landscaping to the greatest practicable extent. Native Florida plants shall be promoted where possible. Where a non-potable water source is available for irrigation, more highly water dependent vegetation may be used.



POLICY 5.1.3.2: DEEP AQUIFER WATER CONSERVATION.

In order to protect the quantity of the Floridan aquifer, the City shall continue to coordinate with the SFWMD and other applicable regulatory agencies in identifying free flowing deep aquifer wells and in requiring corrective measures which control the discharge of water from deep aquifer.

POLICY 5.1.3.3: FUTURE WATER DEMAND.

The City's 10-Year Water Supply Facilities Work Plan shall be reviewed and updated as needed to correspond with the Regional Water Supply Plan, dated 2015. The City's 10-Year Water Supply Facilities Work Plan shall include a minimum 10-year planning period to ensure potable water demands are considered, and groundwater quantity is adequate, to support projected demand. The City shall coordinate with TWA regarding the anticipated water demand and available supply.

POLICY 5.1.3.4: EMERGENCY CONSERVATION OF WATER SOURCES.

The City shall continue to coordinate with the SFWMD in implementing emergency water conservation measures based on the SFWMD plans for management of the region's water resources.

OBJECTIVE 5.1.4: MAINTENANCE OF FLOODPLAIN.

The City shall continue to preserve floodways and prohibit their reduction in the available storage to the floodplain in order to protect and maintain their flood-carrying and flood storage capacity.

POLICY 5.1.4.1: ENFORCE POLICIES TO MAINTAIN FLOODPLAIN.

The City shall continue to provide that flood control measures for new development minimize fill within the 100-year floodplain. All development within the floodplain shall be consistent with all federal, state and local rules.

POLICY 5.1.4.2: MINIMIZE FLOOD DAMAGE.

The City's stormwater management and flood protection policies and regulations shall maintain consistency with program policies of the Federal Insurance Administration. The City shall continue to monitor new cost effective programs for minimizing flood damage. Such programs may include modification in construction setback requirements or other site design techniques, as well as upgrade building and construction techniques to protect against flood hazards.

OBJECTIVE 5.1.5: PROTECT AND PRESERVE WETLANDS.

The City's wetlands shall continue to be protected and preserved from physical and hydrologic alterations.

POLICY 5.1.5.1: WETLAND DEVELOPMENT RESTRICTIONS.

Wetlands (i.e., wetlands shall include hydric soils and wetland species identified by the Florida Department of Environmental Protection (FDEP)) shall be protected from physical or hydrologic alterations in order to maintain natural functions. Development in designated wetlands shall be limited to passive recreation, open space, restricted accessway, bird sanctuary, natural stormwater retention/detention and natural preserve, in accordance with the City's Conservation Overlay regulations in the Land Development Code. At a minimum, encroachment into a designated wetland shall be limited by the 50 foot wetland buffer, and shall be mitigated in accordance with current South Florida Water Management District and FDEP policy.

POLICY 5.1.5.2: LITTORAL ZONES.

A vegetated littoral zone shall be established around functional wetlands occurring on all property, excluding beaches, seawalled areas, and navigation areas.

OBJECTIVE 5.1.6: COMBAT SOIL EROSION.

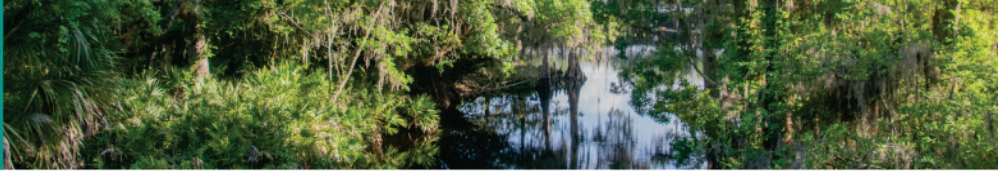
Continue to reduce the incidence of soil erosion caused by land clearing, breaches in stabilized shorelines, and lands having exposed soil without vegetative cover by utilizing best management practices identified by the Florida Department of Environmental Protection.

POLICY 5.1.6.1: IMPLEMENTING EROSION CONTROL.

The City shall continue to specify appropriate measures to be taken during land clearing and building operations to assure that exposed, de-stabilized or otherwise altered soil is expeditiously covered with an acceptable erosion control material. These provisions shall be applicable to the act of subdividing and installation of related improvements as well as during the development review process including the period during which improvements are being undertaken. Tree protection and erosion control measures shall be applicable to all clearing and grading activities and shall include specifications for managing vegetation and re-vegetation.

OBJECTIVE 5.1.7: PREVENTING POTENTIAL ADVERSE IMPACTS OF FUTURE MINING AND EXCAVATION ACTIVITIES.

No mining activities shall be permitted within the City of Kissimmee due to the presence of natural systems that would potentially receive irretrievable losses from the impacts of such operations.



POLICY 5.1.7.1: PROHIBITION AGAINST MINING ACTIVITIES.

Land development regulations shall prohibit mining activities based on the irretrievable losses which such intense activities may potentially impose on the City's ecosystem

POLICY 5.1.7.2: RECLAMATION ACTIVITIES.

The City shall require that mining sites, including borrow pits, located within unincorporated Osceola County undergo reclamation as a condition of voluntary annexation. The procedures and specifications for achieving such reclamation shall continue to be included in the land development regulations.

OBJECTIVE 5.1.8: PROTECT NATIVE VEGETATION.

The City shall continue to protect and retain native vegetation, including hardwood swamps and freshwater marshes, particularly the hardwood hammock community and wetlands along the shoreline of Lake Tohopekaliga, Shingle Creek, Mill Slough, and the City Ditch.

POLICY 5.1.8.1: IMPLEMENTING PROTECTION OF VEGETATIVE COMMUNITIES.

The City shall continue to maintain tree protection and land clearing regulations, as well as landscape regulations and shoreline protection rules, which shall be used in managing and protecting the impacts of development on hardwood swamps and freshwater marshes. The regulations shall mandate fair and equitable restoration and/or compensatory mitigative measures in order to compensate for loss of vegetation and to enhance stabilization of fragile slopes and/or shorelines.

OBJECTIVE 5.1.9: PROTECTING FISHERIES, WILDLIFE AND WILDLIFE HABITATS AND OTHER ENVIRONMENTALLY SENSITIVE AREAS.

The City shall continue to coordinate with Osceola County, the South Florida Water Management District (SFWMD), and the State in protecting fisheries, wildlife, and wildlife habitat.

POLICY 5.1.9.1: MANAGE THE IMPACTS OF DEVELOPMENT ON FISHERIES.

The City of Kissimmee shall continue to coordinate with the SFWMD, the Florida Fish and Wildlife Conservation Commission, and the Florida Department of Environmental Protection, as appropriate, in reviewing the implications of development proposals, including proposed subdivisions and site plan review petitions. Such coordination shall be designed to assist in identifying potential adverse impacts of proposed development on fisheries. The City shall continue to enforce performance standards that are designed to preserve the water quality within Lake Tohopekaliga, and which protect aquatic grassbeds and in order to preserve aquatic habitats and fisheries.



POLICY 5.1.9.2: PROTECT WILDLIFE AND WILDLIFE HABITATS.

The City shall continue to enforce provisions that restrict development activities known to adversely impact endangered, threatened, or rare wildlife and wildlife habitats as well as wildlife and wildlife habitats of special concern. The City shall continue to further protect wildlife and wildlife habitats by use of conservation easements.

POLICY 5.1.9.3: DESIGNATION OF NATURAL RESERVATIONS.

Natural reservations shall be designated as Conservation land uses on the Future Land Use Map.

OBJECTIVE 5.1.10: HAZARDOUS WASTE MANAGEMENT.

The City of Kissimmee shall continue to coordinate with Osceola County as well as appropriate Federal, State and regional agencies for managing hazardous waste.

POLICY 5.1.10.1: MANAGING HAZARDOUS WASTE.

The City shall adhere to the provisions set forth in the Resource Conservation and Recovery Act (RCRA) (U.S. Congress, 1976) and its amendments in order to protect potable water supplies, environmentally sensitive soils, native vegetation, and air quality from the harmful effects of hazardous wastes. The City shall continue to work with the County and appropriate State and regional agencies in identifying industries which use hazardous materials or generate hazardous wastes.

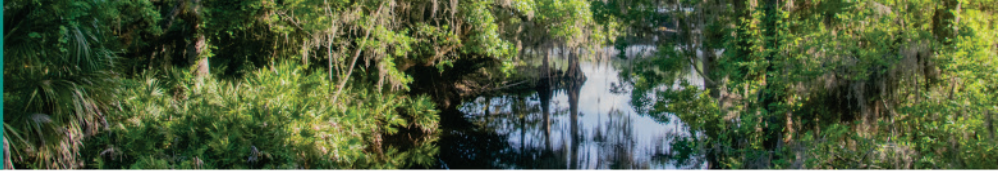
OBJECTIVE 5.1.11: ENCOURAGE THE CONSERVATION AND APPROPRIATE USE OF ENERGY.

The City of Kissimmee shall continue to monitor state-of-the-art concepts for managing energy resource conservation and reducing greenhouse gas emissions. Consider application of those concepts that have been demonstrated to be successful and cost effective in resolving development and conservation issues.

POLICY 5.1.11.1: ENERGY SUPPLY.

Promote an adequate, flexible, reliable, economical and environmentally sound future supply of energy. The City of Kissimmee shall continue to:

- Assist in developing and improving the availability of accurate cost-effective information concerning energy supplies available to the region and County.
- Assist in promoting diversification of energy sources by encouraging a safe and orderly transition from diminishing petroleum resources to alternative, and/or cleaner burning energy technologies as they become available.



- Promote orderly transition from present non-renewable fuels to renewable energy sources and consumption patterns.
- Work with the State, East Central Florida Regional Planning Council (ECFRPC) and Osceola County to promote use of demonstrated cost efficient energy technology employing resource recovery systems and recycling efforts which reduce greenhouse gas emissions and total energy use, reducing depletion of natural resources.

POLICY 5.1.11.2: USE OF ENERGY.

The City shall continue to:

- Consider the effectiveness of total energy use by promoting practices that are cost effective and less energy-intensive.
- Promote practices that ensure that each form of energy is used to do work for which it is best suited
- Recognize the energy services provided by natural features of the environment, including landscape, sun and wind, and promote site development and resource management that complements or substitutes for energy-intensive technologies.
- Encourage citizens to undertake individual and group actions to conserve energy and other resources and assist in providing knowledge of techniques for achieving this objective.

POLICY 5.1.11.3: MANAGEMENT OF ENERGY SUPPLIES AND USE.

The City shall continue to:

- Promote management of energy supplies and use which is consistent with environmental quality, health, safety, social, and economic well-being of the public.
- Minimize the emission of greenhouse gases by encouraging the appropriate use of energy.
- Recognize and assist in meeting the minimum energy needs of the City.
- Assist in minimizing the environmental, economic, and social impacts of future energy and energy-related facilities within the City.

POLICY 5.1.11.4: ENERGY EFFICIENT LAND DEVELOPMENT.

The City shall continue to:

- Encourage energy efficient land development.
- Promote management and development of physical, natural, economic, and human resources in ways that avoid unnecessary long- term energy-intensive investments.
- Recognize the relative energy dependency of commercial and industrial land uses and consider energy dependency in any policy pertaining to new industry promotion strategies or policy concerning maintenance or expansion of existing industry or commerce.
- Encourage land use patterns that by location, scale, and design minimize long-term energy commitments to construction, operation, maintenance, and replacement.
- Encourage natural resource conservation and use in ways that are consistent with sound energy management principles.

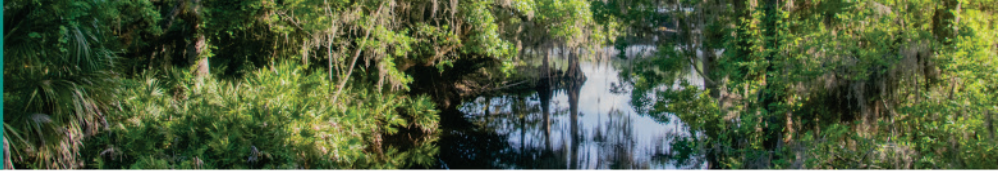
POLICY 5.1.11.5: ENERGY EFFICIENT LAND USE AND TRANSPORTATION.

The City shall continue to implement and enhance multimodal transportation networks through the Multimodal Transportation District, the citywide Transportation Concurrency Exception Area, and the land use plans and policies encouraging mixed-use, pedestrian-oriented development, with higher densities and intensities at locations served by transit, or transit-supportive areas, to conserve energy use and reduce greenhouse gas emissions.

POLICY 5.1.11.6: COORDINATION OF ENERGY MANAGEMENT.

The City shall continue to:

- Coordinate energy management with concerned entities within the public and private sectors.
- Coordinate formulation of energy related decisions with concerned federal, state, regional, and County agencies as well as with concerned private entities.
- Work with these agencies and entities in order to maximize awareness of energy related problems, and issues, alternative techniques for resolving energy related problems and issues, and to identify future areas where joint efforts may enhance mutual goals and objectives.



POLICY 5.1.11.7: ENERGY EFFICIENCY IN PLANS.

The City shall continue to:

- Encourage land use, transportation systems, and urban design that minimizes energy consumption and maximizes effectiveness of energy consumed.
- Reduce travel demands by locating major traffic generators on accessible sites situated along major corridors near potential users.
- Promote a systematic approach to the development of pedestrian, bicycle path, and transit networks by the public and private sectors in order to improve energy efficient transportation links between major activity areas such as residential neighborhoods, employment centers, shopping areas, parks, and schools.

POLICY 5.1.11.8: ENERGY EFFICIENT DESIGN.

The City shall continue to:

- Promote site planning and design which reduces demand for artificial heating, cooling, ventilation and lighting.
- Design factors include building design, siting and orientation that effectively utilize natural solar resources, wind conditions, tree canopy, and plant material to reduce the effects of exposure to extreme weather conditions.

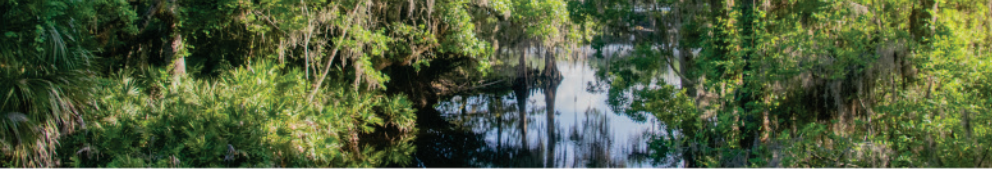
Energy efficient construction shall be promoted through enforcement of the building and energy codes, through application of new and proven energy-efficient technology and through cooperative efforts with building trades, design professionals, building officials, and county, regional and state agencies concerned with energy conservation.

POLICY 5.1.11.9: ENERGY CONSERVATION IN BUILDING AND CONSTRUCTION.

The City shall continue to enforce energy efficient building codes and promote efficient energy conservation in building heating and cooling systems. The City should promote training workshops in energy efficiency in construction and continue to foster cooperative relationships between building trades, architects, engineers and building officials.

POLICY 5.1.11.10: MONITORING NEW ENERGY CONSERVATION TECHNIQUES.

The City shall continue to monitor and evaluate new cost effective techniques for managing land development, energy conservation, and greenhouse gas emissions. The City shall coordinate these reviews with the State and the ECFRPC. These review efforts shall consider innovation in analysis of energy supplies; alternative energy



sources; energy consumption patterns; cost implications; and energy related impacts of utilities including the electrical utility, water and wastewater systems, and solid waste disposal. Energy use in housing, transportation, industry and commerce shall be monitored and evaluated on a continuing basis using available analytical techniques. Analytical findings shall be used to formulate public policy directed toward needed corrective energy conservation measures.

POLICY 5.1.11.11: MUNICIPAL ENERGY-EFFICIENCY.

The City shall continue to lead by example and incorporate energy-efficient designs and strategies in municipal buildings and facilities in order to encourage the construction, and retrofiting, of energy-efficient structures in the City.

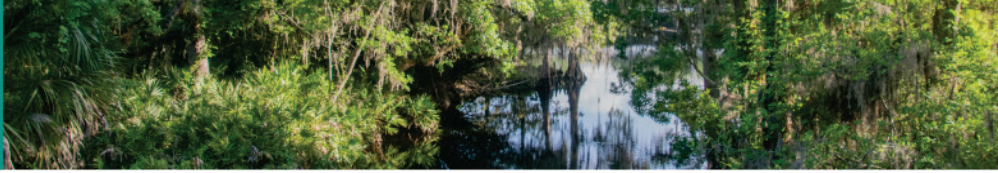
OBJECTIVE 5.1.12: INTERGOVERNMENTAL COORDINATION FOR MANAGING CONSERVATION ACTIVITIES.

The City shall maintain an intergovernmental coordination mechanism to manage natural resources and assist in implementing appropriate laws, ordinances, and plans of existing State, regional and local agencies sharing responsibilities for managing natural resources within the City.

POLICY 5.1.12.1: IMPLEMENTING POLICIES FOR INTERGOVERNMENTAL COORDINATION IN MANAGING CONSERVATION ACTIVITIES.

The City of Kissimmee shall continue to coordinate with Osceola County and appropriate regional, State, and federal agencies in managing conservation resources. Management activities shall include, but not be limited to the following:

1. The City shall coordinate the following conservation resource management issues with Osceola County and other public entities as needed:
 - a. Participate in County forums and in technical review activities concerning emergency preparedness, directed toward issues including update of the Peacetime Emergency Preparedness Plan.
 - b. Participate in technical review concerning water quality, floodplain management, and surface water management, and fish and wildlife, vegetative communities, especially issues impacting major environmentally sensitive resources such as Lake Tohopekaliga, Shingle Creek, and Mill Slough. The activities shall also be coordinated with State and regional entities having appropriate jurisdictional interests.
 - c. Public access issues be coordinated with Osceola County, including issues surrounding maintenance and improvements.



2. Major issues to be coordinated with the Florida Department of Environmental Protection (FDEP) and the South Florida Water Management District (SFWMD) as may be appropriate in managing the following activities:
 - a. The City shall coordinate all development and resource conservation measures impacting Lake Tohopekaliga with the Florida Fish and Wildlife Conservation Commission and FDEP, as well as other applicable public agencies. These activities shall include, but not be limited to, review of proposed development potentially impacting natural resources, including development petitions for docks, shoreline stabilization, dredging, or other alteration of natural resources under the State's jurisdiction.
 - b. The City shall coordinate with technical staff within the SFWMD, and FDEP, in order to assure implementation of sound principles and practices of conservation resource management during the development review process as well as in the formulation of policies impacting coastal resource management.
 - c. The City shall coordinate with the SFWMD as well as other appropriate State agencies in matters surrounding stormwater management, floodplain protection, drainage, water quality and quantity, and consumptive use permitting.

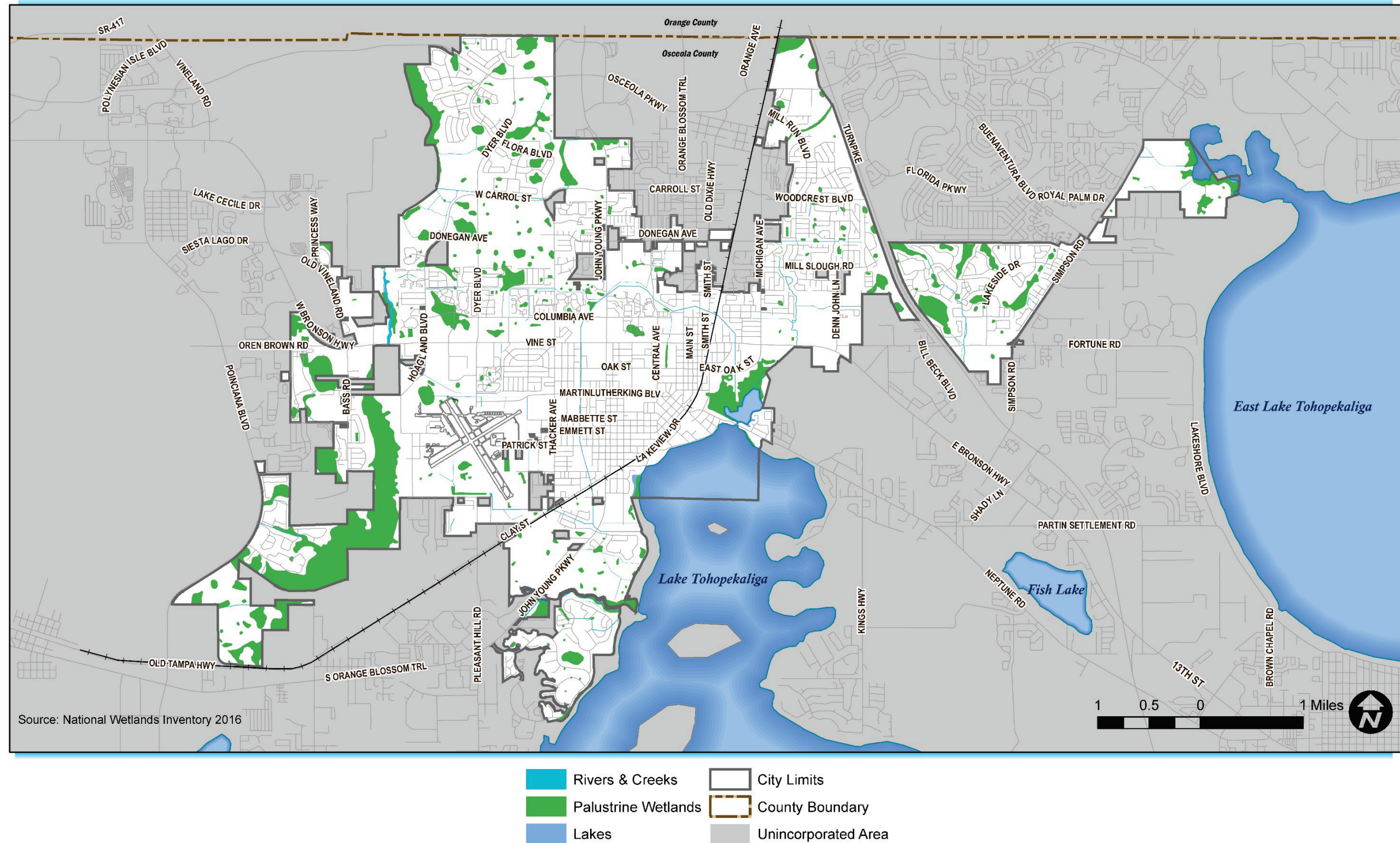


MAPS

- Wetlands and Water Bodies**
- Wells Area**
- Flood Plains**
- Soil Types**
- Topography**
- Vegetative Cover**
- Rare Species Habitat Map**

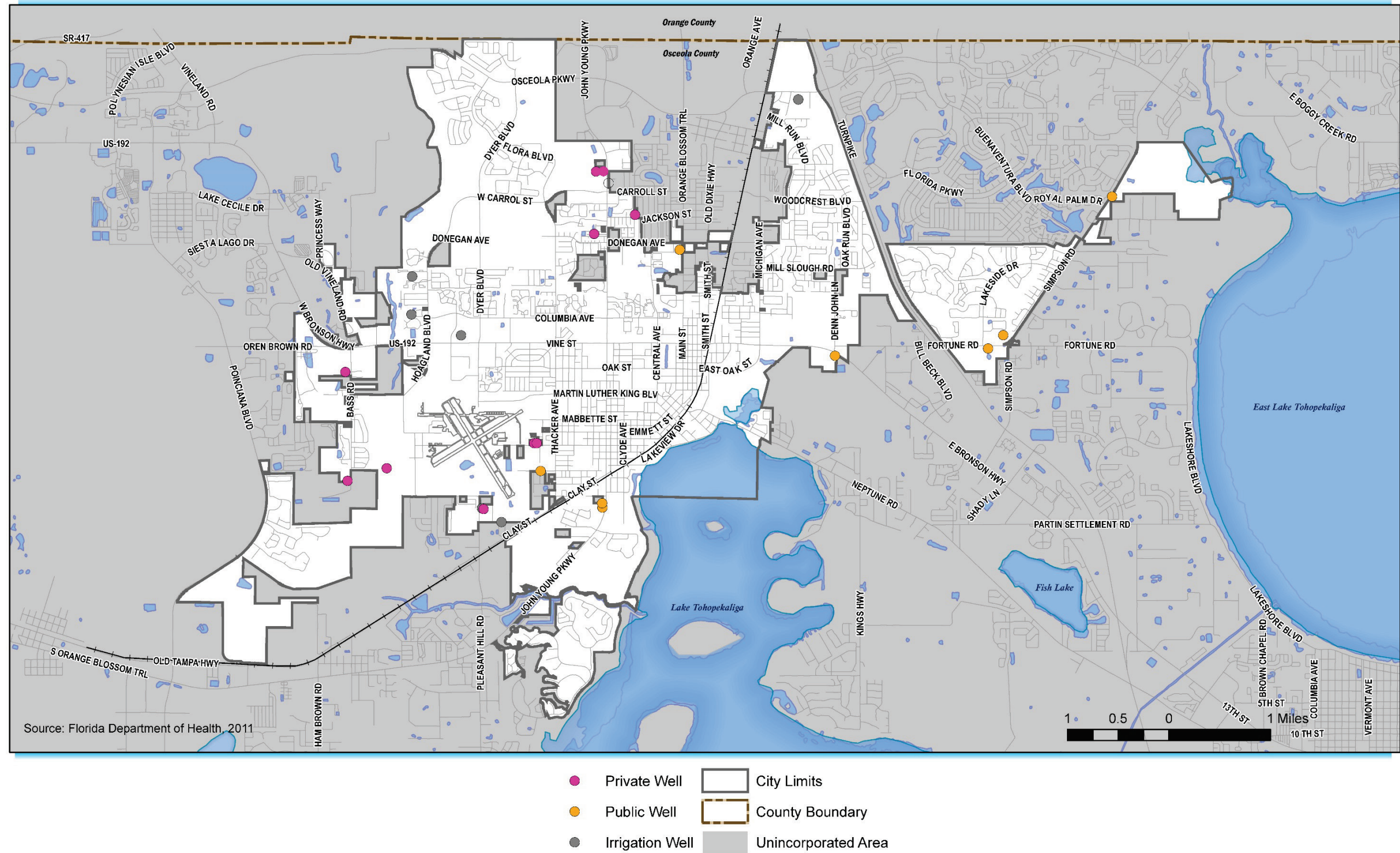
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Figure 5-1: Wetlands and Water Bodies Map



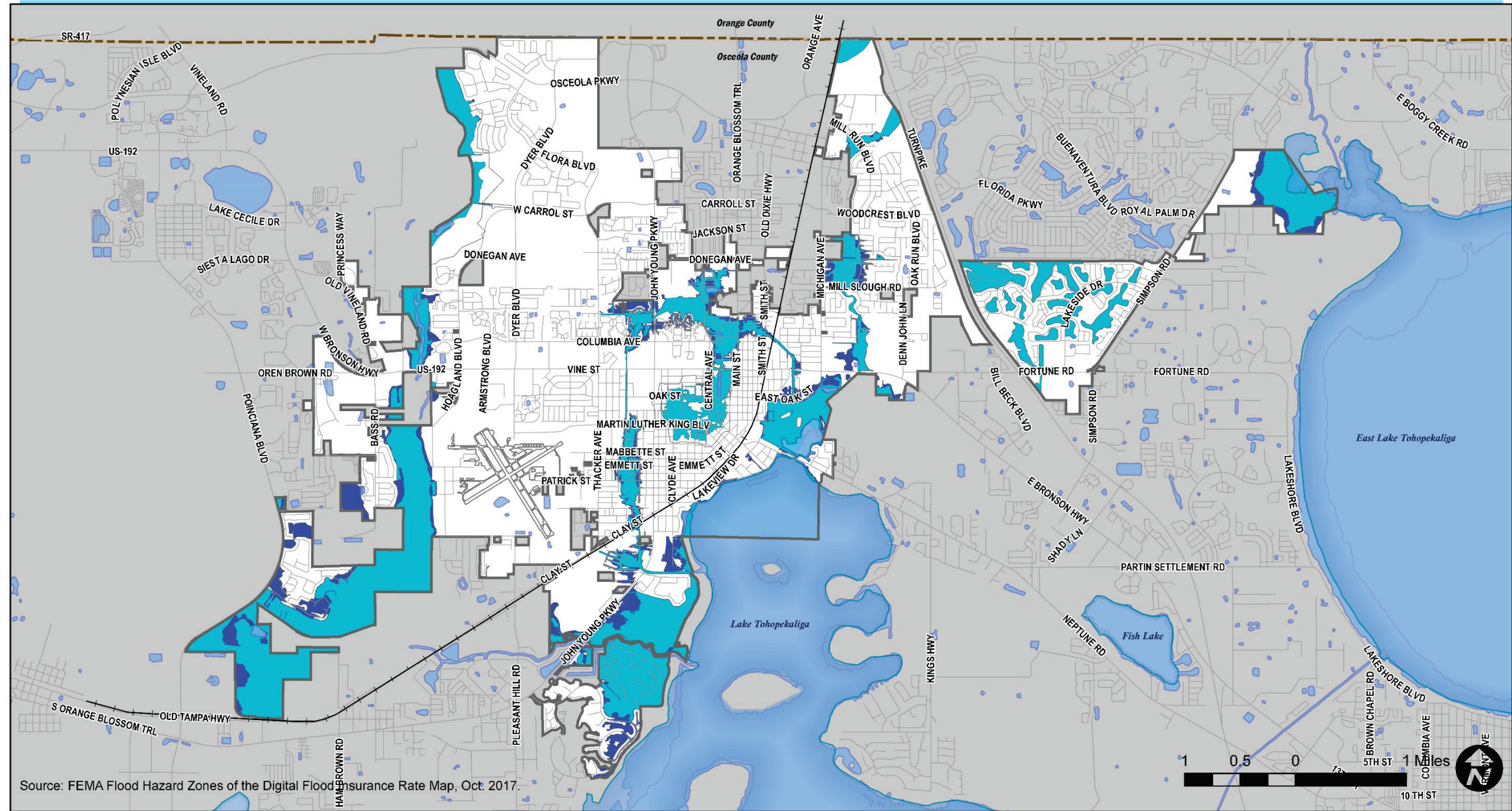
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Figure 5-2: Wells Area Map



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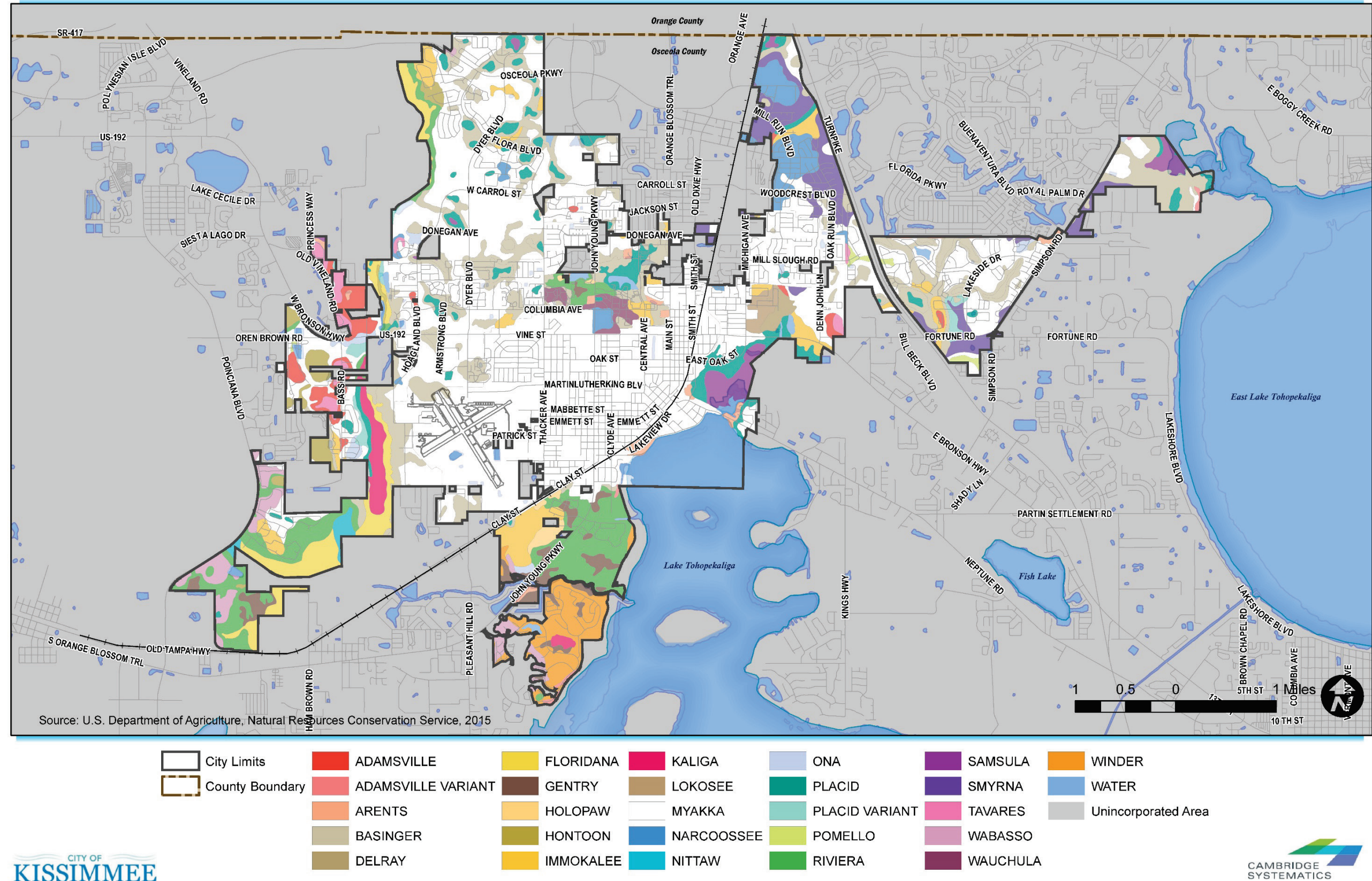
Figure 5-3: Flood Plains Map



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|---------------------|---------------------|
| FLOODPLAIN | City Limits |
| 100-YEAR FLOODPLAIN | County Boundary |
| 500-YEAR FLOODPLAIN | Lake |
| OUTSIDE FLOODPLAIN | Unincorporated Area |

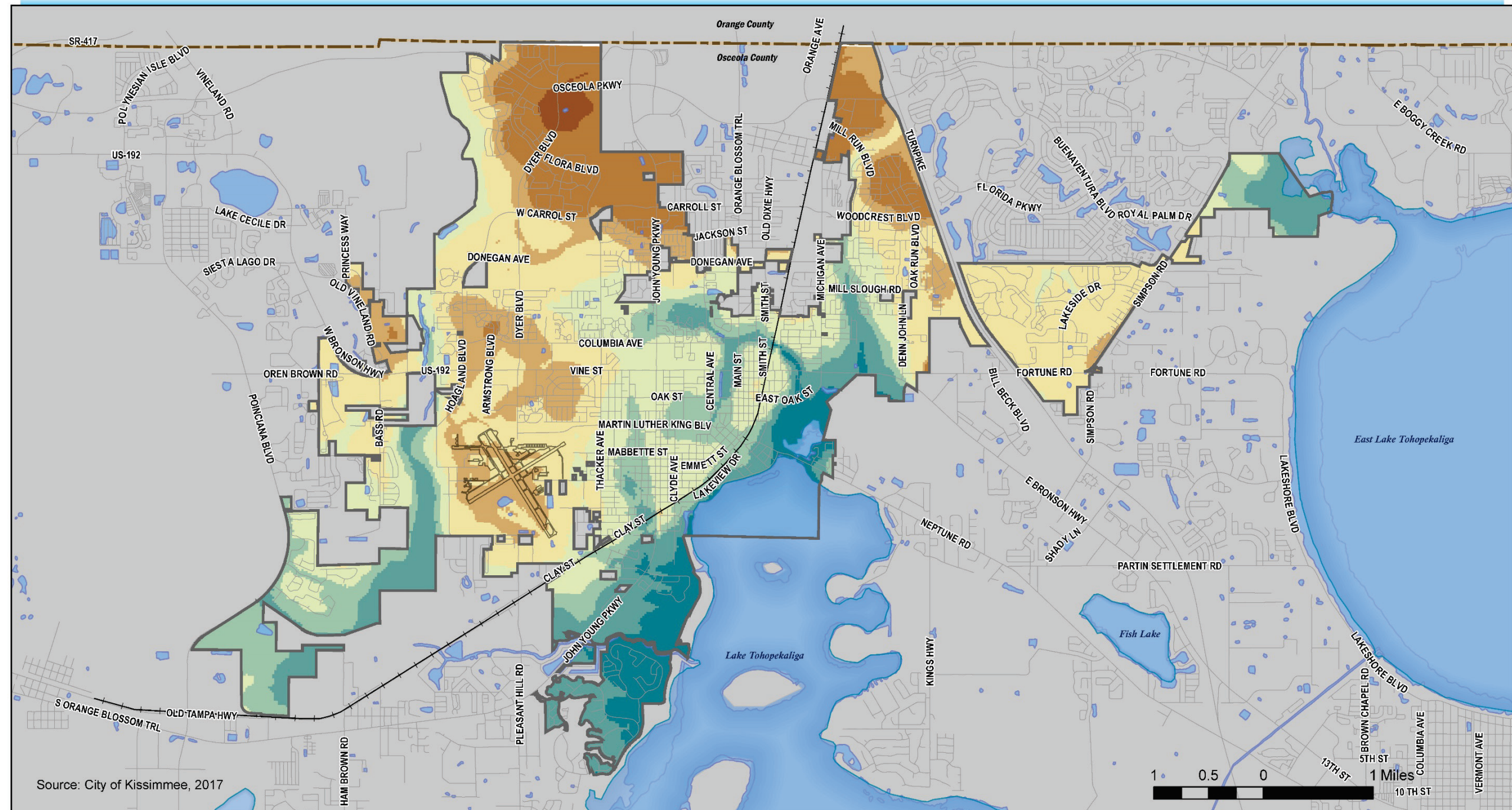
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Figure 5-4: Soil Types Map

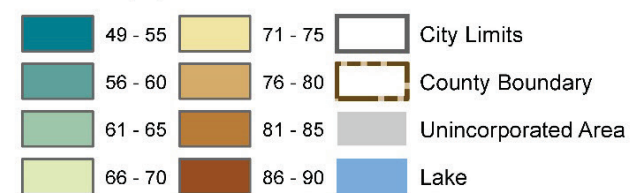


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Figure 5-5: Topography Map

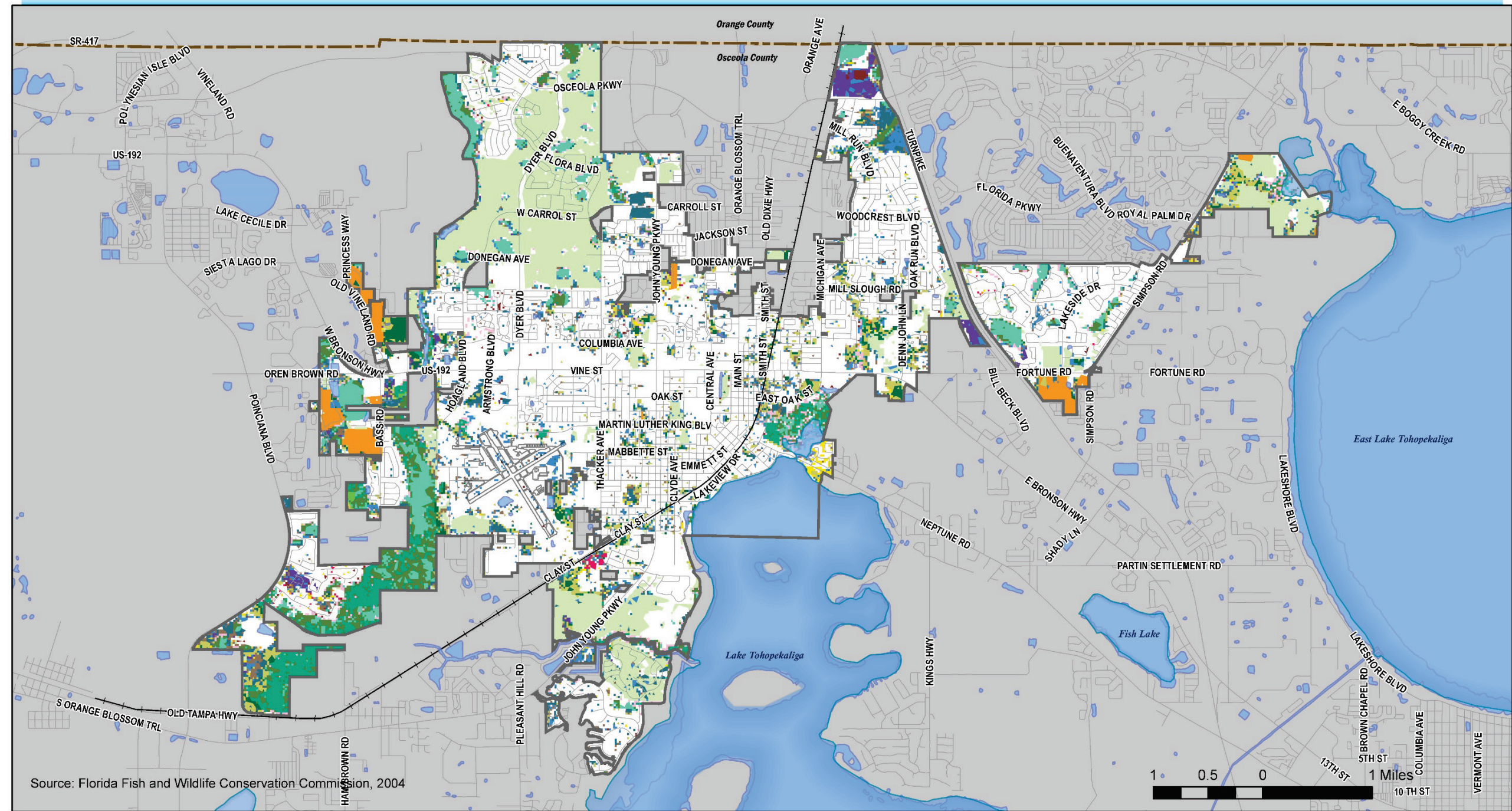


Elevation (ft.)



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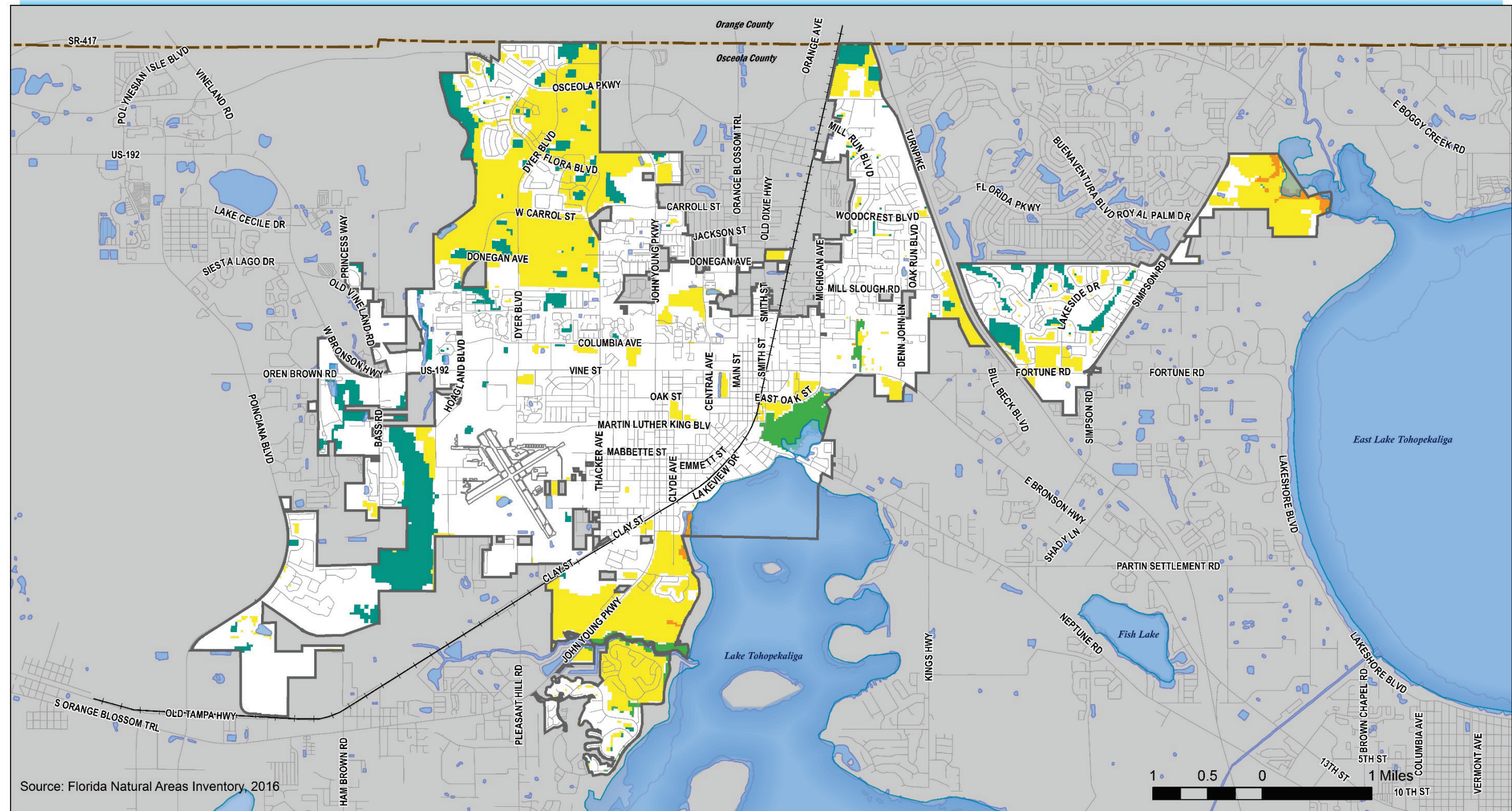
Figure 5-6: Vegetative Cover Map



- | | | | | |
|--|---|---|---|---|
| Bare soil/Clearcut | Dry Prairies | Improved Pasture | Pinelands | Unimproved/Woodland Pasture |
| Bay Swamp | Grassland | Mixed Hardwood-Pine Forests | Row/Field Crops | City Limits |
| Citrus | Hardwood Hammocks and Forests | Mixed Wetland Forest | Shrub Swamp | County Boundary |
| Cypress Swamp | Hardwood Swamp | Other Agriculture | Shrub and Brushland | Unincorporated Area |

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Figure 5-7: Rare Species Habitat Map



Rare Species Habitat Conservation Priorities
(Priority 1 - highest, Priority 6 - lowest)



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DATA, INVENTORY, AND ANALYSIS

INTRODUCTION

The natural environment in and around the City of Kissimmee has been a draw to residents and visitors since the City was founded. This Element provides descriptions of the primary characteristics and qualities of this environment, including impacts, and efforts to conserve and manage the City's natural resources and energy resources.

WATER RESOURCES

The City of Kissimmee is located in the Kissimmee River Basin, on the northern shore of Lake Tohopekaliga, which is part of the Upper Kissimmee River Chain of Lakes. These waters flow all the way from the headwaters of Shingle Creek southward to Lake Okeechobee, the Everglades, and finally into Florida Bay. The lakes, creeks, and associated wetland systems provide a variety of economic, recreational, and wildlife benefits, including habitat for fish and wildlife, and nutrient removal from stormwater runoff. The locations of these waterbodies, and wetlands, are shown in *Figure 5-1: Wetlands and Water Bodies Map*.

There are two major natural watersheds in the City. The Shingle Creek watershed flows south from Orange County, and runs parallel to the west side of the Airport before turning eastward to flow into Lake Tohopekaliga. The Lake Tohopekaliga watershed is a major Kissimmee River Basin sub-basin, and part of the Kissimmee River Chain of Lakes. The Lake receives inflow from other major surface water systems including Shingle Creek, Mill Slough, Canoe Creek, Reedy Creek, and the East Lake Tohopekaliga watershed.

SURFACE WATER QUALITY AND QUANTITY

There are three water bodies within the City of Kissimmee that are identified as Class III water bodies, these include: Lake Tohopekaliga, Shingle Creek, and Mill Slough. These water bodies provide recreational opportunities to the Kissimmee community and attract visitors that bolster the local economy. Impacts of concern to these waters include erosion, runoff from nearby agricultural uses and minor point source pollutants such as automotive repair shops.

LAKE TOHOPEKALIGA

The Lake has had a history of poor water quality resulting from high levels of nitrogen and phosphorous resulting from the use of fertilizers and agricultural waste products. These sources include:



Former nearby citrus groves and dairy farms;

- Past effluent discharges from sewer treatment plants located in Osceola County and Orange County; and
- Urban stormwater run-off from the City of Kissimmee and surrounding areas.

As noted in the Public Facilities Element, the water quality of the Lake has improved over the past two decades. The Lake does, however, still have problems with the accumulation of mercury in the fish population. The EPA has approved the Total Maximum Daily Load (TMDL) submitted by FDEP, a plan to restore impaired waters by limiting the amount of a pollutant that enters a waterbody.



The Lake also has problems with invasive exotic aquatic plants, such as *hydrilla*, and was listed on the State impaired waters list due to nutrients from an imbalance of flora and fauna. The City and its partners have submitted a Nutrient Reduction Plan (NRP) to FDEP to improve and monitor water quality without a TMDL. The SFWMD, with their jurisdictional control of the Lake, is involved in aquatic weed control programs on the Lake.

SHINGLE CREEK

As with Lake Tohopekaliga, past water quality reports from Shingle Creek indicated high concentrations of nutrients, primarily caused by effluent discharged from the McLeod Road and Sand Lake Road wastewater treatment plants. Both plants have ceased discharging sewage into the Creek, and no other point sources of pollution are discharging into the Creek. The Creek is still impacted by stormwater runoff from both the City, and areas in Orange County. Shingle Creek is on the FDEP list of impaired waters and was identified as a medium priority for TMDL development. Mill Slough

This waterway also has its headwaters in southern Orange County, and flows southward through the City and into Lake Tohopekaliga on the eastern side of downtown. Pollution enters the Slough primarily from stormwater runoff, as well as nearby irrigation systems.

The Floridan Aquifer, underlying most of the state of Florida and all of Osceola County, is the primary source of water supply for the City.

POTABLE WATER SUPPLY AND DEMAND

The Floridan Aquifer, underlying most of the state of Florida and all of Osceola County, is the primary source of water supply in the City. The Public Facilities Element Chapter 4A: Potable Water Sub-Element.

WATER CONSERVATION

Voluntary residential and commercial water conservation will be achieved through the City's participation in water conservation efforts, such as promoting the use of water-saving plumbing devices including low-flow toilets, showerheads, and faucets within new developments, as well the development and expansion of reclaimed water systems. The City's adoption of the Work Plan also included the adoption of several policies addressing the conservation of water supplies and reduction of water demands. Conservation measures that may be incorporated into the City's Land Development Code include:

- Restricting landscape irrigation to two days a week;
- Utilizing Florida Water Star criteria for the construction of new single-family residences;
- Limiting the use of St. Augustine turf to front yards and entries;
- Requiring the construction of dual distribution systems in all developments to provide for non-potable irrigation;
- Prohibiting the use of potable water for irrigation in new developments while requiring the use of reclaimed water, stormwater and other sources of non-potable water; and
- Prohibiting construction of irrigation wells into the Floridan aquifer.

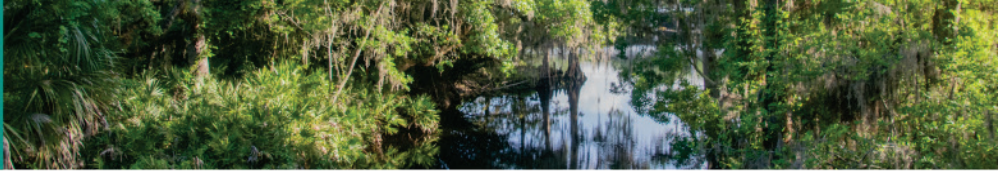
Additional information regarding potable water supply conservation measures can be found in Chapter 4A.

REUSE

Water reuse in central Florida is accomplished primarily through the use of reclaimed water for irrigation. Reclaimed water is treated wastewater that has received at least secondary treatment and basic disinfection, and is reused after flowing out of a domestic wastewater treatment facility. Reuse is the deliberate application of reclaimed water for a beneficial purpose in compliance with the rules of FDEP and the SFWMD. Reclaimed water strategies in the regional water supply plans can include such measures as development of urban reclaimed water systems, reclaimed water system interconnections, supplemental supplies for peak flows, and aquifer storage and recovery. TWA provides reclaimed water for non-potable uses such as landscape irrigation and as cooling water for an electrical generating power plant.

GROUNDWATER

As noted in the Aquifer Recharge Sub-Element (Chapter 4C), the City is underlain by both the Surficial and Floridan aquifers. The Floridan is the source of current water supplies for all the City's potable water



supplies. *Figure 5-2: Wells Area Map* shows the locations of public and private water supply wells, and areas of different rates of groundwater recharge.

POTENTIAL GROUNDWATER CONTAMINATION SITES

There are two locations in or near the City that may pose a problem for the contamination of groundwater by hazardous chemicals, and both are in the Shingle Creek watershed. The first is the Animal Diagnostic Lab on John Young Parkway, which in the past discharged certain solid wastes to a septic tank/drainfield system, resulting in localized contamination of the Surficial Aquifer. This site is located within one block of a major Floridan Aquifer wellfield for public supply, but recharge to that aquifer is not likely. However, the potential exists for migration of contaminated shallow groundwater into either the drinking water supply, or through groundwater discharge to swales and stormwater facilities.

The second site is the Bass Road Resource Recovery Site, which operated as a Class I facility and accepted residential, commercial, industrial, and agricultural wastes. It is primarily a recycling center and offers disposal of tires, household chemicals, and yard waste as well. It is located within one mile of a Floridan Aquifer wellfield. Osceola County is actively engaged in monitoring activities at this site.

FLOODPLAINS

The National Flood Insurance Program administered by the Federal Emergency Management Agency has determined that the bottomlands adjacent to Shingle Creek, Mill Slough, and Lake Tohopekaliga are subject to flooding and inundation from 100-year storm events, or a one percent chance of occurring annually.

The National Flood Insurance Program administered by the Federal Emergency Management Agency has determined that the bottomlands adjacent to Shingle Creek, Mill Slough, and Lake Tohopekaliga are subject to flooding and inundation from 100-year storm events.

The extent of the 100-year floodplain is shown in *Figure 5-3: Flood Plains Map*. Other floodplain locations in the City include the City Ditches, the Medical District area, and north of Vine Street between John Young Parkway and Central Avenue. Areas within the 100-year flood zone may experience flooding during less severe storms. Prior to development within floodplains, measures are taken to assure that the natural storage areas are preserved or that compensating storage is provided in a development. Floodplain protection and development regulations are included in the Land Development Regulations.

SOILS AND GEOLOGY

MAJOR SOIL TYPES

There are numerous types of soils under the City, though as noted, many are poorly drained and have limitations for use as septic drainfields. *Figure 5-4: Soil Types Map* identifies the location of the soils present within the City. The three most prevalent soil types within the City of Kissimmee are Myakka, Basinger, and Riviera.

MYAKKA

Covering approximately 6,700 acres, the Myakka series consists of deep and very deep, poorly drained to very poorly drained soils that formed in sandy marine deposits. These soils are in areas of flatwoods, in high tidal areas, on floodplains, or in depressions. They have rapid permeability in the A horizon and moderate or moderately rapid permeability in the Bh horizon. Slopes range from 0 to 8 percent.

BASINGER

These soils cover approximately 1,600 acres, and consist of very deep, poorly drained and very poorly drained, rapidly permeable soils in sloughs, depressions, low flats, and poorly defined drainageways. They formed in sandy marine sediments. Slopes range from 0 to 2 percent.

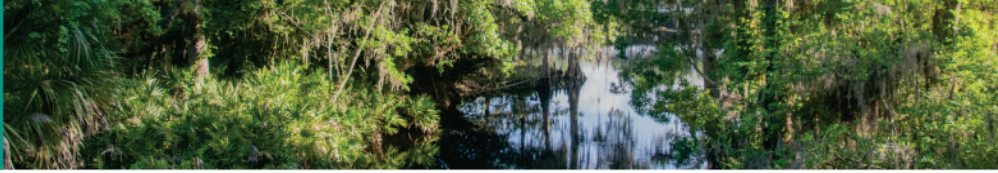
RIVIERA

The Riviera soil series covers approximately 795 acres in the City. The series consists of very deep, poorly drained, very slowly permeable soils on broad, low flats and in depressions in the Lower Coastal Plain. They formed in stratified sandy and loamy marine sediments, and slopes range from 0 to 2 percent.

SOIL EROSION

According to the Soil Conservation Service, soil erosion is not a significant problem in any specific areas within or adjacent to the City. The level, poorly drained soils found throughout the City are not conducive to wind or water erosion. General locations that may be conducive to erosion include canal areas and construction sites. The banks of canals may erode, causing alluvial sediment to build up on canal floors and reducing the carrying capacity of the drainage-ways. This erosion is frequently caused by runoff from canals or from construction activities. Construction sites also contribute to erosion through clearing and vegetation removal activities.

The FDEP requires construction sites to be permitted through the NPDES, in order to ensure reductions in the potential for stormwater runoff and the resulting erosion. The use of construction best



management practices, in addition to stormwater management best management practices, will help control stormwater runoff within the City, particularly at development sites.

MINERAL RESOURCES

There are no commercially exploited mineral resources within the City of Kissimmee. Most of the major mining operations and commercially valuable mineral deposits within the region are located to the southwest in Polk County.

TOPOGRAPHY

Elevation in the City of Kissimmee ranges from 49-90 feet above sea level. The highest elevations are generally in the northern portions of the City, furthest from its water bodies. The lowest elevations are adjacent to Lake Tohopekaliga and East Lake Tohopekaliga, as well as the Palustrine Wetlands in the southwest. *Figure 5-5: Topography Map* shows the elevation throughout the City.

HAZARDOUS WASTES

There are currently no active landfills within the limits of the City of Kissimmee. Steps shall be taken to continue to monitor those inactive sites within the city and any problems related to hazardous waste through coordination with Florida Department of Environmental Protection and the Florida Bureau of Hazardous Waste. There are also two landfills within in Osceola County adjacent to the City's Southwest border and several inactive landfills within the City. There are no brownfield sites identified in Kissimmee on the Florida Department of Environmental Protection Brownfield Redevelopment Program. There are currently no sites within the City of Kissimmee identified on the Environmental Protection Agency's Superfund National Priorities List.

VEGETATIVE COMMUNITIES

An illustration of the locations of particular types of vegetative communities, including wetlands, is shown in *Figure 5-6: Vegetative Cover Map*. The majority of the City is either a developed urban area or an improved pasture, which is slowly evolving to urban development, primarily residential uses.

LISTED SPECIES AND HABITAT PROTECTION

The City has been established for over 130 years, and development has steadily occurred since then. There has, however, been far more development in the past 50 years than in the years prior to then. This last 50 years also happens to coincide with the growth in awareness of the natural environment and the need to maintain certain habitats for the species that live within them. *Figure 5-7: Rare Species Habitat Map* identifies the locations that are most likely to support listed species. Although there are no Priority 1 or 2 areas within the City, and only a smattering of Priority 3 areas near Lake Tohopekaliga, there are large swaths of Priority 4 habitat conservation areas in the Shingle Creek drainage adjacent to Lake Tohopekaliga, as well as in the area between Columbia Avenue and Osceola Parkway.



Table 5-1 profiles the listed species in Osceola County as determined at the state or federal level. New development within the City must adhere to state legislation regarding these species and minimizing impacts to their habitats.

Table 5-1: Listed Species in Osceola County

Scientific Name	Common Name	Federal Classification*	State Classification*
Plants and Lichens			
<i>Andropogon arctatus</i>	pinewoods bluestem	N	T
<i>Bonamia grandiflora</i>	Florida bonamia	T	E
<i>Calopogon multiflorus</i>	many-flowered grass-pink	N	T
<i>Carex chapmannii</i>	Chapman's sedge	N	T
<i>Chionanthus pygmaeus</i>	pygmy fringe tree	E	E
<i>Coleataenia abscissa</i>	cutthroatgrass	N	E
<i>Conradina grandiflora</i>	large-flowered rosemary	N	T
<i>Eriogonum longifolium</i> var. <i>gnaphalifolium</i>	scrub buckwheat	T	E
<i>Gymnopogon chapmanianus</i>	Chapman's skeletongrass	N	N
<i>Lechea cernua</i>	nodding pinweed	N	T
<i>Lechea divaricata</i>	pine pinweed	N	E



Scientific Name	Common Name	Federal Classification*	State Classification*
<i>Lupinus aridorum</i>	scrub lupine	E	E
<i>Nemastylis floridana</i>	celestial lily	N	E
<i>Nolina atopocarpa</i>	Florida beargrass	N	T
<i>Nolina brittoniana</i>	Britton's beargrass	E	E
<i>Ophioglossum palmatum</i>	hand fern	N	E
<i>Paronychia chartacea</i> var. <i>chartacea</i>	paper-like nailwort	T	E
<i>Pecluma plumula</i>	plume polypody	N	E
<i>Pecluma ptilota</i> var. <i>bourgeauana</i>	comb polypody	N	E
<i>Polygala lewtonii</i>	Lewton's polygala	E	E
<i>Polygonella myriophylla</i>	Small's jointweed	E	E
<i>Prunus geniculata</i>	scrub plum	E	E
<i>Pteroglossaspis ecristata</i>	giant orchid	N	T
<i>Schizachyrium niveum</i>	scrub bluestem	N	E
<i>Warea amplexifolia</i>	clasping warea	E	E
Amphibians			
<i>Lithobates capito</i>	Gopher Frog	N	N
<i>Notophthalmus perstriatus</i>	Striped Newt	C	N
Reptiles			
<i>Alligator mississippiensis</i>	American Alligator	SAT	SAT
<i>Crotalus adamanteus</i>	Eastern Diamondback Rattlesnake	N	N
<i>Drymarchon couperi</i>	Eastern Indigo Snake	T	T
<i>Gopherus polyphemus</i>	Gopher Tortoise	C	T
<i>Lampropeltis calligaster</i>	Mole Kingsnake	N	N
<i>Lampropeltis getula</i>	Common Kingsnake	N	N
<i>Pituophis melanoleucus</i>	Pine Snake	N	T
<i>Plestiodon egregius lividus</i>	Blue-tailed Mole Skink	T	T
<i>Plestiodon reynoldsi</i>	Sand Skink	T	T
<i>Sceloporus woodi</i>	Florida Scrub Lizard	N	N
Birds			
<i>Ammodramus savannarum floridanus</i>	Florida Grasshopper Sparrow	E	E
<i>Antigone canadensis pratensis</i>	Florida Sandhill Crane	N	T
<i>Aphelocoma coerulescens</i>	Florida Scrub-Jay	T	T
<i>Aramus guarauna</i>	Limpkin	N	N
<i>Athene cunicularia floridana</i>	Florida Burrowing Owl	N	T
<i>Buteo brachyurus</i>	Short-tailed Hawk	N	N
<i>Caracara cheriway</i>	Crested Caracara	T	T
<i>Egretta caerulea</i>	Little Blue Heron	N	T
<i>Egretta thula</i>	Snowy Egret	N	N
<i>Egretta tricolor</i>	Tricolored Heron	N	T

Scientific Name	Common Name	Federal Classification*	State Classification*
<i>Elanoides forficatus</i>	Swallow-tailed Kite	N	N
<i>Elanus leucurus</i>	White-tailed Kite	N	N
<i>Eudocimus albus</i>	White Ibis	N	N
<i>Falco columbarius</i>	Merlin	N	N
<i>Falco peregrinus</i>	Peregrine Falcon	N	N
<i>Falco sparverius paulus</i>	Southeastern American Kestrel	N	T
<i>Grus americana</i>	Whooping Crane	XN	XN
<i>Haliaeetus leucocephalus</i>	Bald Eagle	N	N
<i>Laterallus jamaicensis</i>	Black Rail	N	N
<i>Mycteria americana</i>	Wood Stork	T	T
<i>Nyctanassa violacea</i>	Yellow-crowned Night-heron	N	N
<i>Nycticorax nycticorax</i>	Black-crowned Night-heron	N	N
<i>Peucaea aestivalis</i>	Bachman's Sparrow	N	N
<i>Picoides borealis</i>	Red-cockaded Woodpecker	E	E
<i>Picoides villosus</i>	Hairy Woodpecker	N	N
<i>Plegadis falcinellus</i>	Glossy Ibis	N	N
<i>Rostrhamus sociabilis</i>	Snail Kite	E	E
Mammals			
<i>Corynorhinus rafinesquii</i>	Rafinesque's Big-eared Bat	N	N
<i>Eumops floridanus</i>	Florida bonneted bat	E	E
<i>Mustela frenata peninsulae</i>	Florida Long-tailed Weasel	N	N
<i>Neofiber alleni</i>	Round-tailed Muskrat	N	N
<i>Podomys floridanus</i>	Florida Mouse	N	N
<i>Puma concolor coryi</i>	Florida Panther	E	E
<i>Sciurus niger shermani</i>	Sherman's Fox Squirrel	N	SSC
<i>Ursus americanus floridanus</i>	Florida Black Bear	N	N

Source: Florida Natural Areas Inventory (FNAI), 2018

* Federal and State Legal Status: Provided by FNAI for information only. Consult the relevant federal agency, for official definitions and lists of protected species.

C=Candidate species for which federal listing agencies have sufficient information on biological vulnerability and threats to support proposing to list the species as Endangered or Threatened

E=Endangered

N=Not listed

SAT=Endangered due to similarity of appearance to a species which is federally listed

SSC=Species of Special Concern by Florida Fish and Wildlife Conservation Commission

T=Threatened

XN=Non-essential experimental population



USE AND PRESERVATION OF NATURAL RESOURCES AND COMMUNITIES

Though farming and ranching have been important to the City's history, the City no longer maintains large agricultural or ranching operations, as are found in other parts of Osceola County. However, some transitioning agricultural land uses are present in the north and south edges of the City. The predominant uses of natural resources in the City are for recreation and conservation. There are no commercial operations in the City that directly use natural resources, such as for mining, commercial fisheries, or timber.

RECREATION

The Recreation and Open Space Element identifies the City's parks, recreation areas and open spaces. Additionally, the City's location on Lake Tohopekaliga, and the numerous lakes surrounding the City, makes the area an attractive location for fishing, boating, and other water-dependent uses. Lake Tohopekaliga constitutes a substantial recreational resource, and is a popular fishing destination. The most popular sport fish harvested from Lake Tohopekaliga, and other nearby lakes, include largemouth bass, bluegill, black crappie, red-ear sunfish (shell cracker), brown bullhead (catfish), and spotted sunfish (stumpknocker).

CONSERVATION

The City, through the Conservation future land use designation, among other means, engages in the conservation and preservation of natural resources, including the protection of Shingle Creek through land acquisition, and the promotion of open space protection. Other conservation activities include the protection of groundwater and surface water resources through stormwater best management practices, low-impact development techniques, and through water conservation.

AIR QUALITY

The quality of ambient air, or outside air, is monitored by the FDEP. Air-quality monitoring stations are placed in areas where the ambient air-quality standards are being violated, or in areas where standards are being met but plans are needed to ensure maintenance of acceptable levels in the face of anticipated population or industrial growth.

The City has one ozone monitoring station, located at the Four Corners Fire Station. This site began monitoring ozone levels in 1993.



The State of Florida including the City of Kissimmee are in attainment for Particulate standards (<2.5 micrometers and <10 micrometers) as set by the EPA. Stations in the City do not measure for other pollutants, such as carbon monoxide, lead, nitrogen dioxide, or sulfur.

As of 2015, the three year running average for ozone was 63 parts per billion (ppb). The non-attainment threshold for ozone levels is 75 ppb, indicating the City is in attainment for existing air quality ozone standards.

Ground-level ozone is a pollutant, and is not to be confused with atmospheric ozone, which is necessary to block harmful radiation from the sun. Ground-level ozone is formed from reactions between nitrogen oxides and volatile organic compounds (VOCs) in the presence of sunlight. VOCs are typically generated by motor vehicles, but may also arise from the evaporation of gasoline and solvents, open burning, or other combustion sources. The worst ozone conditions occur on hot, calm days.

ENERGY EFFICIENCY

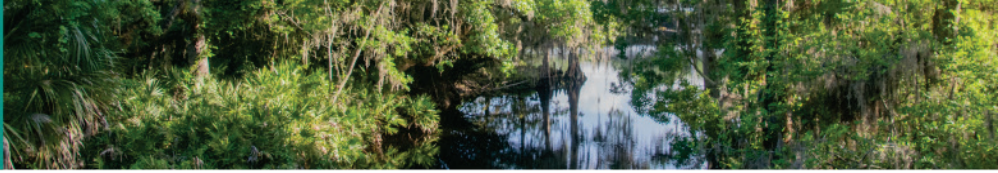
The City, through the Comprehensive Plan's adopted goals, objectives and policies, has for years demonstrated a consideration of issues such as energy use and greenhouse gas emissions, long before they were mandated by the Florida Legislature. Existing policy topics include management of energy supplies and use, energy efficient land development, energy efficiency in plans, and energy conservation in building and construction.

The Transportation Element incorporates a Multi-Modal Transportation District (MMTD), and multi-modal supportive policies. The Land Use Element also incorporates increased densities within the MMTD. These policies encourage transit use, and walking and bicycling in what may lead to a more compact urban form as more full-time residents in the MMTD area draw businesses, and promote additional compact development in a sustainable cycle, rather than developing linearly along roadway corridors. The Comprehensive Plan also contains policies that encourage the preservation of wetlands for natural stormwater attenuation.

To reduce energy use and greenhouse gas (GHG) emissions, the City of Kissimmee has identified areas within the City where existing development patterns are most energy efficient. These areas are compact allowing for a denser network of transportation and utilities infrastructure with greater connectivity, which require less energy input to provide service to residents and businesses.

The MMTD encompasses areas of the City with the most efficient and integrated transportation networks. The Community Redevelopment Area (CRA), which is inside the MMTD area, includes the older, densest portion of Kissimmee including the historic downtown.

The City also is setting citywide and transit station-specific policies to support and encourage the use of commuter rail and bus transit.



Additionally, the City of Kissimmee recognizes the importance of natural systems. Trees and vegetation clean the air of pollutants, and sequester greenhouse gasses through transpiration. More wildlife increases biodiversity helping to support rare and unique species that attract ecotourists, and wetlands provide retention and filtration of stormwater runoff. The use of low-impact development techniques introduce vegetation to help filter stormwater, and a test project initiated by the City involves the use of “permeable paving” for parking lots, pathways, and secondary roads when applicable. Permeable pavement is an environmentally friendly, pervious road surfacing mixture. The use of such innovations allows the City to reduce the footprint needed for stormwater facilities, while also contributing to groundwater recharge.

To more effectively meet these natural resource conservation ends, the City has partnered with Osceola County and the SFWMD to purchase and preserve Shingle Creek and adjacent lands, and maintains a series of parks and wetlands to maintain open space and ecological and biological functions throughout the City. These services would be costly for the City to replace with constructed facilities, which would require additional energy inputs.

SUMMARY

The preservation, conservation, and management of natural systems within the City will ensure future generations are able to experience wildlife habitats, clean water resources, and clean air. The built environment has a tremendous impact on human health, the environment, and the economy, and patterns of development implemented today will impact the way future generations interact with their environment. The City will remain proactive, and continue to work to ensure residents and visitors are mindful of the consumption of energy and the creation of greenhouse gasses, and that everyone must work together to provide for the City’s future.